# A PALEOMAGNETIC STUDY OF BASALT FLOWS FROM THE ABSAROKA MOUNTAINS, WYOMING

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Abstract. A paleomagnetic study of 34 flows from the Absaroka Volcanic Supergroup (Eocene) in northwestern Wyoming showed that 23 of these flows have normal polarity, 7 have reversed polarity, and 4 are anomalously magnetized. Two of the anomalously magnetized flows occur between normally and reversely magnetized flows and may have been extruded during a field reversal; the other two are probably contaminated with a stable lightning-induced remanence. The magnetic component of the flows is a titanomagnetite in the range 25--40 mol % ulvospinel which has in some cases been oxidized at low temperature to a cation deficient spinel. The pattern of polarities of the flows, which were sampled at four separate locations, provides an excellent basis for refining the correlation between the flows in this area. The pattern also suggests that the 34 flows were extruded over a period of time less than (and perhaps much less than) one million years in length. Nineteen flows satisfied sampling and stability criteria and were used to determine a new Eocene pole position for North America at 83.5°N, 177.4°E with an $\alpha_{95}$ of 10.1°. Paleosecular variation of the magnetic pole during this part of the Eocene was about 17°.

## Introduction

During the summers of 1970 and 1973 a total of 34 volcanic flows from four separate locations in the Absaroka Mountains of northwest Wyoming were sampled for a paleomagnetic study. The primary objective of this program was to establish paleomagnetic criteria with which to correlate sequences of flows belonging to the Absaroka Volcanic Supergroup [Smedes and Prostka, 1972]. These flows were extruded in the Eocene Epoch [Nelson and Pierce, 1968; Smedes and Prostka, 1972], a time when the earth's field was reversing frequently [Heirtzler et al., 1968]. Both normal and reversed polarities are present in the sampled flows, thus providing an excellent basis for correlation. The study also provides valuable new information about the behavior of the magnetic field during this time, since there have been relatively few paleomagnetic studies of Eocene rocks from North America. As a result the North American pole position for the Eocene is poorly substantiated, and apart from the Siletz River volcanics [Cox, 1957] and the Virginia dikes [Lovlie and Opdyke, 1974], there has been a perplexing absence of reversals reported from North American Eocene volcanic rocks.

The area of this study lies between the town of Cody, Wyoming and the eastern boundary of Yellowstone National Park (see index map, Figure 1). This is a region of relatively flat lying moderately dissected lava flows, breccias, agglomerates, sandstones, and siltstones. The flows belong to two formations of the Absaroka Volcanic Supergroup, the Wapiti Formation and the overlapping Trout Peak Trachyandesite [Smedes and Prostka, 1972]. The Wapiti Formation is composed chiefly of volcanically derived sediments; almost all the flows in this formation occur in a restricted area and are assigned to the Jim Mountain Member [Nelson and Pierce, 1968]. The Trout Peak Trachyandesite consists predominantly of flows with some intercalated clastic rocks. The source area for both groups of flows was probably to the north of the study area, as is indicated by radiating dike patterns and the fact that both the thickness of the formations and the proportion of volcanic to clastic rocks increase northward. The Wapiti Formation overlies the chiefly sedimentary Willwood Formation; the Trout Peak Trachyandesite is overlain by the volcanic Langford Formation in the Sylvan Pass area and by the volcaniclastic Wiggins Formation on Ptarmigan and Carter mountains [Pierce and Nelson, 1969].

Correlation of all units of the Wapiti Formation and the Trout Peak Trachyandesite has been a very difficult problem. Nelson and Pierce [1968] reported extensive chemical analyses of flows from both formations. They found that virtually all flows were of trachyandesite composition and that there were no chemical means which could be used to unambiguously distinguish one flow from another. Fossil flora and fauna are neither diagnostic nor widespread enough to provide marker horizons in sedimentary units, although they can be used for rough age assignments [Dorf, 1939; Hay, 1956; Pierce, 1963]. The most promising attempts at correlation have been based on recognition of similar relationships between sequences of flows and associated sediments, but these relationships do not provide a firm basis for correlation over even moderate distances because of rapid lithologic changes along strike and because the pattern of extrusive units is a strong function of distance from the source area.

Several potassium-argon ages obtained throughout the Absaroka volcanics are reported by Smedes and Prostka [1972]. One, near Sylvan Pass in the upper Trout Peak Trachyandesite, gives an age of 48.0 ± 1.3 m.y. B.P. (biotite). Other determinations have been made for units stratigraphically above the Trout Peak Trachyandesite and located about 80 km to the south of the study area. The Two Ocean Formation is dated at 48.5 ± 1.3 m.y. B.P. (biotite) and 47.9 ± 1.3 m.y. B.P. (sanidine), the Tepee Trail Formation at 47.8 ±

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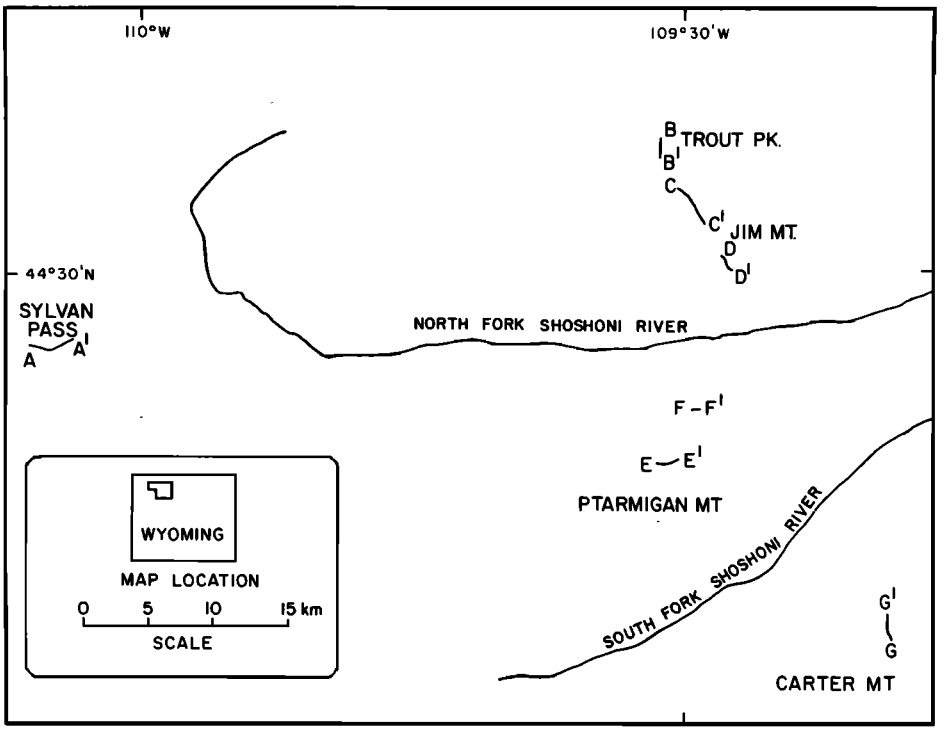



Fig. 1. Index map of sampling locations. The labeled lines (A-A', etc.) refer to the cross sections of Figure 2, where more detailed sampling information is presented.

1.3 m.y. B.P. (sanidine) and 47.9 ± 1.3 m.y. B.P. (biotite), and the Wiggins Formation between 47.1 and 44.4 m.y. B.P. on the basis of seven dates from hornblende and biotite. These dates, together with the paleontological evidence mentioned above, indicate that the flows of this study are definitely of Eocene age.

Very little magnetic work has been done on the Absaroka volcanics. F. J. Vine (personal communication, 1971) sampled the Trout Peak Trachyandesite in a different area and found only normally magnetized rocks. W. F. Hanna (personal communication, 1975) sampled correlative rocks north of Yellowstone National Park with the same result. Parsons et al. [1972] carried out a paleomagnetic study on some breccias in the Sunlight Basin north of the area of this study and concluded, on the basis of a consistent direction of magnetization, that they had been emplaced above the Curie Point.

## Sampling and Measurement Procedures

A total of 34 flows was sampled. The cross sections of each location shown in Figure 2 give detailed information about the position of each flow. Four sampling locations were chosen to include the most accessible and most complete sequences of relatively fresh rock from several places in the section. These sequences were found between Trout Peak and Jim Mountain, on Ptarmigan Mountain and Carter Mountain, and just to the east of Sylvan Pass (see Figure 1). Only flows greater than 1 m. thick and of uncontorted resistant rock were sampled. At the Sylvan Pass site, which was easily accessible by road, three flows were sampled with a portable gasoline-powered drill. Because of the distance from roads, the rugged relief, and the lack of water, oriented block samples were collected at all other localities. Twenty-two flows were sampled between Jim Mountain and Trout Peak, seven on Ptarmigan Mountain, and two on Carter Mountain. Usually, between three and five separate block samples were collected at equal intervals between the top and bottom of each flow. Five flows were sampled at two different locations (at least 0.3 km apart) as a test for consistency. Care was taken to avoid samples clearly affected by lightning strikes, but as most of the collecting was necessarily done on exposed cliffs and riges, it was impossible to avoid such samples completely. The block samples were drilled into 2.4-cm-diameter cores and reoriented in the laboratory. The oriented cores were cut into individual specimens 2.4 cm long; most cores were long enough to yield two to three specimens.

All specimens were measured on a Schonstedt SSM-1 spinner magnetometer to obtain natural remanent magnetization (NRM) values. Most specimens were then subjected to stepwise alternating field demagnetization in peak fields of 100, 150, and 200 Oe, since a pilot program showed that these levels were most likely to be effective. Specimens from a few flows were demagnetized in peak fields as great as 1000 Oe in an attempt to remove obvious secondary components.

Mean directions of magnetization were calculated for each flow at each level of demagnetization. The most representative value of the mean direction for a flow was made by choosing the data set from the demagnetization level for which

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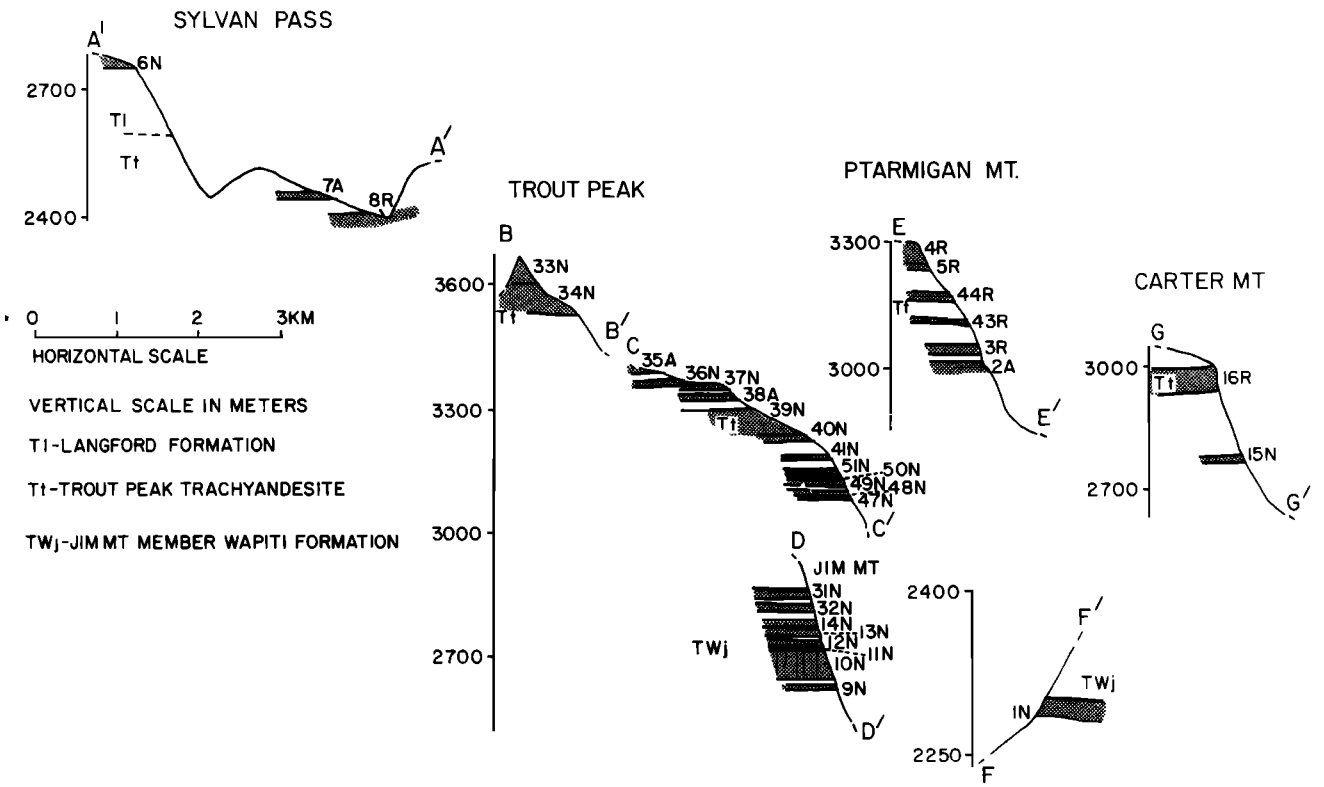



Fig. 2. Cross sections at the individual sampling locations showing the positions of the flows in the sections. The flows are labeled N (normal), R (reversed), or A (anomalous) according to their polarity.

the sample directions were most tightly clustered. In most cases the optimum demagnetization level was 150 Oe; in a few cases the statistics were better at 100 or 200 Oe. Increasing the demagnetization level to improve flow statistics which were poor at the 200-Oe level and below were successful in several cases. No mean directions were calculated from raw NRM data.

Virtual geomagnetic pole (VGP) positions were calculated for each flow, and a mean VGP was determined for the group of flows. To be included in the calculation of the group mean, a flow had to pass the following two criteria: (1) the within-flow statistics for the flow mean direction had to have an $\alpha_{95}$ value less than 20°, and (2) the flow data had to be based on at least three samples. These criteria eliminated 15 flows from consideration. It is very important to note that many flows that were judged unsuitable for inclusion in the calculation of a mean VGP are nevertheless sufficiently stably magnetized to be assigned a polarity.

The mean VGP for all flows meeting the two criteria above was calculated under the assumption that the individual flow VGP's are Fisherian but the individual flow directions may not be [Cox, 1970]. This should be true if the group flows were extruded over an interval of time which is considerably longer than the period of secular variation of the earth's field, which is apparently of the order of 10,000 years [Cox, 1969]. This condition is probably satisfied, as evidence will be presented below which suggests that the flows represent a time interval of the order of one million years. The VGP for the group is thus calculated as the mean of the individual flow VGP's rather than as the VGP corresponding to the mean declination and inclination of the individual flow directions.

## Magnetic Mineralogy

Because of the weathered nature of most of the flows we felt that it was important to determine the composition of the magnetic phases present and the nature of the chemical changes they might have suffered. A recording magnetic balance similar to that described by Cox and Doell [1967] was used to measure saturation magnetization versus temperature during heating and cooling cycles for six whole rock samples. Three samples were taken from flow 44 (Ptarmigan Mountain) which appearred to be highly weathered; the other three were taken from flow 16a, a flow from Carter Mountain appearing relatively unweathered. The approximately 300-mg samples were heated to about 600°C under a vacuum between $1 \times 10^{-4}$ and $4 \times 10^{-4}$ torr in a constant gradient field with an intensity at the sample of 9000 Oe.

The magnetization curves for the six samples are shown in Figure 3. All three samples from flow 44 show Curie points near 500°C on heating, slightly higher Curie points on cooling, and slight irreversibility during the cooling cycle. The samples from flow 16a contain phases with low Curie temperatures (between 200°C and 300°C) in addition to a phase with a Curie temperature of about 500°C on heating. On cooling there is no

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indication of the lower Curie temperature phase, and there is a significant increase in the saturation magnetization.

X ray diffraction studies of magnetic separates and electron probe microanalysis of opaque grains from both flows provided further information about the magnetic minerals. Samples from both flows contain a spinel phase with a unit cell edge of about 8.40 Å; flow 16a contains in addition a spinel with a cell edge of about 8.44 Å. These determinations are approximate, as there was scatter in the cell edge values for each spinel as determined from different reflections. Iron-titanium ratios from microprobe measurements showed that magnetic grains from flow 44 are about 24 mol % ulvospinel (x = 0.24), while those from flow 16a have composition corresponding to x = 0.40.

The reversible thermomagnetic behavior of samples from flow 44 shows that it contains reatively unoxidized titanomagnetite. The Curie points support the probe determinations of composition (x = 0.24); the 8.40-Å cell edge measurement is somewhat small but is also subject to the largest uncertainty. The interpretation of the magnetic mineralogy of flow 16a is more complicated and is based in part on the work of Readman and O'Reilly [1972], who measured Curie temperatures and lattice parameters for synthetic titanomagnetites and cation deficient spinels.

The low Curie temperature (200°-300°C) phase with lattice parameter 8.44 Å present in flow 16a corresponds to a titanomaghemite with x = 0.4 which has been only slightly oxidized (z < 0.2). We will designate this as phase A. The increase in saturation magnetization for these samples is due to inversion of phase A to magnetite-rich spinel. The existence of the phase with 8.40-Å cell edge in flow 16a is suggested only by the X ray diffraction measurements. It could be a highly oxidized titanomaghemite with x = 0.4 resulting from low-temperature oxidation of phase A. This would invert partly to a magnetite-rich spinel [Readman and O'Reilly, 1970] which would be more strongly magnetic than the parent titanomaghemite, but there would be much less of it, and the net change in magnetic intensity could be negligible.

It is interesting that this interpretation is not consistent with the more weathered appearance of flow 44 or with X ray diffraction data of whole rock samples, which show that flow 16a contains analcime but few if any clay minerals, while flow 44 contains montmorillonite but no analcime. Perhaps compositions which are richer in titanium are more susceptible to oxidation. In any case, the important result for this study is that these flows are likely to contain titanomagnetites which have been oxidized.

The obvious question to consider at this point is whether or not a titanomagnetite acquires a stable secondary magnetization during low-temperature oxidation to a cation deficient spinel. Apart from the implications for this study, the question is one of the utmost importance in rock magnetism because of the extremely oxidized nature of many crystalline rocks from the sea floor and because paleomagnetic studies on older continental rocks may also be based on samples containing oxidized titanomagnetites. Unfortunately, there is evidence to support both sides

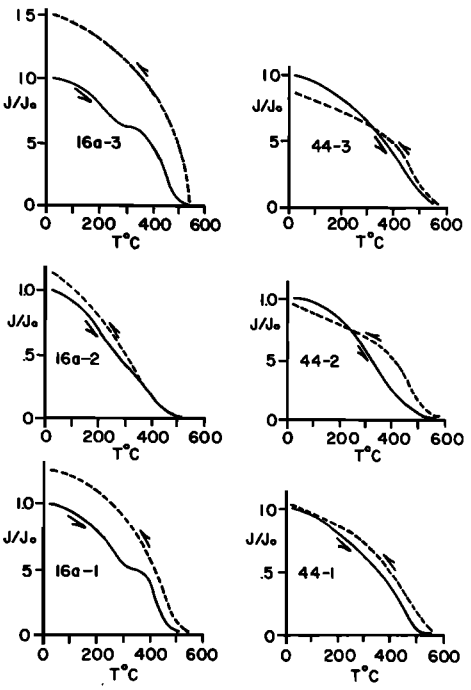



Fig. 3. Saturation magnetization versus temperature for six samples, three from flow 16a and three from flow 44. Heating and cooling were done in vacuum.

of the question. The opposite poles of scientific thought on the matter are probably best illustrated by the work of Johnson and Merrill [1973] and Marshall and Cox [1972]. Johnson and Merrill showed that oxidation of magnetic separates in an oxygen plasma at temperatures between 50° and 195°C produced a chemical remanent magnetization (CRM) in the direction of the field in which oxidation occurred and, even more disturbing, that the CRM was more stable than the original anhysteretic remanent magnetization. Marshall and Cox studied pillows whose rims had undergone natural oxidation but whose centers were fresh and concluded that the oxidized portion was magnetized in the direction of the original thermal remanent magnetization (TRM) and not that of the field during oxidation. More recently, Johnson and Merrill [1974] have shown that single-domain magnetite does not acquire a CRM different in direction from the original remanence during oxidation to maghemite. Unfortunately, we are not able to tell whether or not the carriers of remanence in the Absaroka flows are single domain, although the demagnetization which was done on all samples will certainly discriminate in favor of the single-domain remanence.

In this paper we will operate under the assumption that the flows are recording an Eocene pole position. The factors which lead us to believe that this is so are that (1) the Marshall and Cox experiments were performed on samples exposed to natural weathering conditions, while those of Johnson and Merrill were not, (2) the presence of reversed flows indicates that they could not have acquired substantial CRM during the last 700,000 years, the time during which exhumation and significant weathering must have occurred, (3) the

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TABLE 1. Paleomagnetic Data from Absaroka Flows

| Flow | $H_o$ | N | D | I | $\alpha_{95}$ | δ | κ | λ | θ | Polarity |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Sylvan Pass | | | | | |
| 6 | 100 | 5 | 343.4 | 64.3 | 8.2 | 10.6 | 58.3 | 78.2 | -186.7 | N |
| 7* | 300 | 4 | 336.7 | -87.2 | 23.3 | 27.0 | 9.0 | 39.3 | -107.1 | A |
| 8 | 100 | 5 | 121.8 | -73.1 | 4.9 | 6.3 | 165.6 | 52.6 | -156.7 | R |
| | | | | | Jim Mountain to Trout Peak | | | | | |
| 33* | 150 | 3 | 91.0 | 74.1 | ... | 71.0 | 1.3 | 37.2 | -71.1 | N |
| 34* | 600 | 3 | 60.4 | 48.8 | 31.4 | 31.5 | 6.6 | 40.8 | -23.5 | N |
| 35* | 100 | 3 | 101.4 | -23.6 | ... | 60.4 | 1.8 | 16.7 | -198.6 | A |
| 36* | 300 | 3 | 288.3 | 60.7 | 43.2 | 43.3 | 3.5 | 39.3 | -175.9 | N |
| 37* | 100 | 2 | 18.3 | 51.4 | 13.6 | 11.1 | 53.3 | 71.1 | 15.5 | N |
| 38* | 100 | 3 | 172.7 | 35.7 | ... | 68.5 | 1.4 | -25.3 | -101.9 | A |
| 39 | 200 | 3 | 8.8 | 57.2 | 12.4 | 12.4 | 42.6 | 80.6 | 23.3 | N |
| 40 | 200 | 3 | 319.5 | 36.8 | 12.6 | 12.6 | 41.0 | 48.8 | 138.1 | N |
| 41* | 300 | 3 | 170.9 | 86.1 | 23.2 | 23.3 | 12.1 | 36.8 | -108.0 | N |
| 51 | 100 | 4 | 333.1 | 77.3 | 7.6 | 8.7 | 85.7 | 64.2 | -134.8 | N |
| 50* | 100 | 2 | 344.0 | 82.4 | 4.4 | 3.6 | 501.9 | 58.7 | -117.4 | N |
| 49* | 100 | 4 | 34.1 | 61.0 | 25.7 | 29.8 | 7.4 | 65.3 | -25.9 | N |
| 48 | 100 | 3 | 311.4 | 45.5 | 13.6 | 13.7 | 35.2 | 47.6 | 152.8 | N |
| 47* | 300 | 3 | 27.6 | 60.2 | 29.9 | 30.0 | 7.3 | 69.6 | -19.2 | N |
| 31 | 150 | 3 | 34.6 | 60.1 | 6.7 | 6.7 | 145.0 | 64.6 | -23.9 | N |
| 32 | 150 | 3 | 10.2 | 39.4 | 9.1 | 9.2 | 78.1 | 66.3 | 46.5 | N |
| 14 | 200 | 4 | 349.4 | 61.1 | 9.0 | 10.4 | 60.5 | 81.9 | 147.3 | N |
| 13* | 150 | 2 | 358.9 | 29.9 | 29.1 | 25.1 | 11.6 | 61.5 | 72.8 | N |
| 12 | 150 | 4 | 24.1 | 62.7 | 14.5 | 17.7 | 23.3 | 72.8 | -26.6 | N |
| 11 | 150 | 3 | 13.6 | 52.1 | 6.5 | 6.4 | 155.8 | 74.1 | 24.0 | N |
| 10 | 150 | 5 | 17.1 | 80.1 | 6.0 | 8.5 | 108.0 | 65.1 | -100.2 | N |
| 9 | 200 | 4 | 316.1 | 74.1 | 4.7 | 5.7 | 220.9 | 59.7 | -152.4 | N |
| | | | | | Ptarmigan Mountain | | | | | |
| 4b | 300 | 7 | 165.2 | -68.0 | 22.4 | 38.9 | 5.6 | 78.2 | -161.0 | R |
| 4a | 150 | 3 | 134.4 | -66.7 | 5.6 | 5.6 | 205.5 | 59.0 | -174.2 | R |
| 4 | 300 | 12 | 164.4 | -69.2 | 19.6 | 43.3 | 3.5 | 76.8 | -154.9 | R |
| 5b | 300 | 4 | 137.4 | -11.3 | 13.0 | 15.7 | 29.2 | 36.3 | 127.2 | R |
| 5a | 100 | 5 | 160.1 | -21.1 | 18.5 | 26.0 | 11.5 | 52.3 | 103.6 | R |
| 5 | 300 | 9 | 152.7 | -17.1 | 12.7 | 25.3 | 13.5 | 47.2 | 112.2 | R |
| 44 | 150 | 3 | 161.5 | -69.0 | 11.4 | 11.2 | 50.7 | 75.5 | -161.7 | R |
| 43* | 100 | 2 | 114.6 | -43.5 | 22.4 | 18.1 | 19.6 | 34.7 | -196.6 | R |
| 3b | 100 | 2 | 191.3 | -51.9 | 22.4 | 15.7 | 19.6 | 75.4 | 29.9 | R |
| 3a | 150 | 3 | 152.0 | -78.8 | 4.0 | 4.0 | 415.4 | 62.1 | -131.2 | R |
| 3 | 100 | 5 | 183.0 | -73.3 | 13.2 | 24.9 | 22.6 | 75.3 | -103.4 | R |
| 2b | 150 | 1 | 283.7 | -43.4 | ... | ... | ... | 8.5 | -46.9 | A |
| 2a | 300 | 3 | 359.1 | -16.1 | 17.2 | 17.0 | 22.1 | -37.3 | -108.4 | A |
| 2* | 300 | 4 | 344.1 | -23.4 | 31.1 | 38.3 | 5.0 | -31.5 | -91.2 | A |
| 1 | 200 | 5 | 34.3 | 70.3 | 9.6 | 13.5 | 42.2 | 65.9 | -55.9 | N |
| | | | | | Carter Mountain | | | | | |
| 16b | 200 | 2 | 166.9 | -57.1 | 4.0 | 2.8 | 600.3 | 78.2 | 131.8 | R |
| 16a | 200 | 6 | 172.0 | -31.6 | 10.2 | 16.0 | 31.6 | 61.8 | 86.9 | R |
| 16 | 200 | 8 | 171.1 | -38.1 | 10.4 | 19.3 | 22.7 | 65.7 | 91.1 | R |
| 15* | 200 | 3 | 3.5 | 21.8 | 20.1 | 22.8 | 12.4 | 56.9 | 64.5 | N |
| | | | | | Average for Stable Flows | | | | | |
| Normal | | 13 | 355.1 | 63.2 | 9.4 | 19.7 | 16.9 | | | N |
| Reversed | | 6 | 160.0 | -58.1 | 17.5 | 24.8 | 10.7 | | | R |
| Total | | 19 | 349.9 | 61.8 | 8.4 | 21.1 | 14.7 | | | both |

*Not included in statistical analyses.

pattern of normal and reversed sequences of flows suggests that the remanence is not controlled by weathering, (4) polarities of the more weathered margins of the flows did not disagree with those of the fresher interiors, and (5) polarities of stable flows sampled at different locations did not disagree. We realize that the validity of this assumption rests in part on experiments which have not yet been performed. We should also point out that the low-temperature oxidation of a titanomagnetite of less than 50 mol % ulvospinel should produce a decrease in intensity of magnetization but not a self-reversal [Readman and O'Reilly, 1972].

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## Results

Table 1 presents a summary of the paleomagnetic results from all individual flows sampled in this study. The flow numbers appear in the first column, and these are arranged to match the actual stratigraphic position of the flows at each site. Asterisks indicate flows which were not included in statistical analyses. The next two columns list $H_o$, the value of the optimum demagnetizing field, and N, the number of samples used in the statistical calculations for that flow. This may be smaller than the number of samples collected, since some were eliminated because of secondary (mainly lightning-induced) magnetization. The next seven columns contain D and I (the mean declination and inclination of the flow $\alpha_{95}$, $\delta$ (standard deviation), and $\kappa$ (precision parameter of Fisherian distribution), and $\lambda$ and $\theta$ (latitude and longitude of the VGP for the flow, positive values for $\theta$ indicating east longitude). The final column contains N (normal) R (reversed), or A (anomalous) as an indication of the polarity of the flow. Flows 2, 3, 4, 5, and 16 were sampled at two locations; the table lists data for each location separately in addition to data for samples from both locations combined to give a flow average.

If we temporarily omit from consideration the four flows (2, 7, 35, and 38) whose directions are anomalous, the pattern of polarities for the four sites is fairly simple. At Carter Mountain a reversed flow overlies a normal flow. Except for one normal flow well below the others, the entire sequence at Ptarmigan Mountain is reversed. The section from Jim Mountain to Trout Peak is completely normal, and at Sylvan Pass a normal flow overlies a reversed flow.

Of the four flows whose polarities are uncertain (anomalous) the two within the Jim Mountain-Trout Peak section (35 and 38) have very high NRM intensities and large values of $\alpha_{95}$ and were probably strongly affected by lightning. The other two (2 and 7) cannot be explained so easily. First, their statistics show only slightly more scatter in direction than the average. Second, the directions for flow 2 are anomalous in the same sense (magnetization upward and in the northwest quadrant) at the two locations, a feature which lightning-induced magnetization would be unlikely to produce. Also their NRM intensities are average or below average. A possible explanation is suggested by the observation that the flows above and below these two anomalous flows have opposite polarities. Perhaps the earth's field was reversed during the extrusion of flows 2 and 7, and they are recording a true intermediate field direction.

There are two lines of evidence which indicate that lightning is the probably chief cause of scatter in the data. In most cases where the statistical data for a flow are poor we observed that one or more samples from that flow had a very high value (as high as 0.5 emu/cc) of NRM intensity. Such high intensities are almost invariably the result of lightning strikes [Irving, 1964]. In addition, in the sequence between Jim Mountain and Trout Peak, flows 33-38 show consistently large values of scatter and very high NRM intensities (between 0.06 and 0.18 emu/cc). These flows were the most exposed of any sampled by this study; although great care was taken to collect from areas unaffected by lightning, it was apparently not possible to avoid it for the above six flows.

## Discussion

Figure 4a shows the correlation established by Nelson and Pierce [1968] between three of the four sampling locations of our study. The Nelson and Pierce correlation is based largely on relationships between sequences of flows and sedimentary units and can be refined by the consideration of flow polarity. First, the correlation of the sequence of flows of the Jim Mountain Member at the type section on Jim Mountain to a single flow on Ptarmigan Mountain, 11 km to the south, is strengthened by the magnetic data. All eight flows of the Jim Mountain Member were normally magnetized; the lowest flow at Ptarmigan Mountain is also normally magnetized and is in fact the only flow with normal polarity at that location. This correlation is especially significant because of the marked thinning of the sequence (250 m to 20 m) in such a short distance.

The most serious problem with the correlation of Figure 4a is that the 12 flows from the Trout Peak Trachyandesite between Jim Mountain and Trout Peak are normally magnetized, while all 5 flows from the sequence on Ptarmigan Mountain to which they supposedly correspond have reversed polarity. Because of the evidence indicating that the source area was probably to the north of the study area, it is reasonable to assume that the normal sequence of flows extruded on Trout Peak did not extend to Ptarmigan Mountain. If we use the data that the volcanic section thins to the south away from the source area, we are then bound to assume that the 300-m-thick sequence of reversed flows which once covered Trout Peak has since been eroded away. Although there is no supporting evidence for this hypothesis, we will incorporate it into our correlation on the (perhaps questionable) basis that it is less artificial than assuming the existence of local or multiple volcanic source areas. In any case it is clear that the Trout Peak and Ptarmigan Mountain flows cannot be the same.

Figure 4b shows the revised correlation of this study. We feel that the reversed flow at Carter Mountain corresponds to the reversed sequence on Ptarmigan Mountain and that the normal flow on Carter Mountain probably belongs to the Wapiti Formation (perhaps the Jim Mountain Member). The lowermost flow at Sylvan Pass is reversed and may correlate with the reversed sequence on Ptarmigan Mountain, although the 50-km distance between the two sites leaves many other possibilities open.

The sequence of polarities observed in this study may be compared with the geomagnetic polarity history for the Eocene as extrapolated from sea floor linear magnetic anomalies. The radiometric ages cited above indicate that the flows were extruded sometime between 45 and 50 m.y. B. P., and the sequence of flow polarities suggests that this time included the end of a normal polarity interval, a complete reversed interval, and the beginning of another normal interval. The time scale of Heirtzler et al. [1968]

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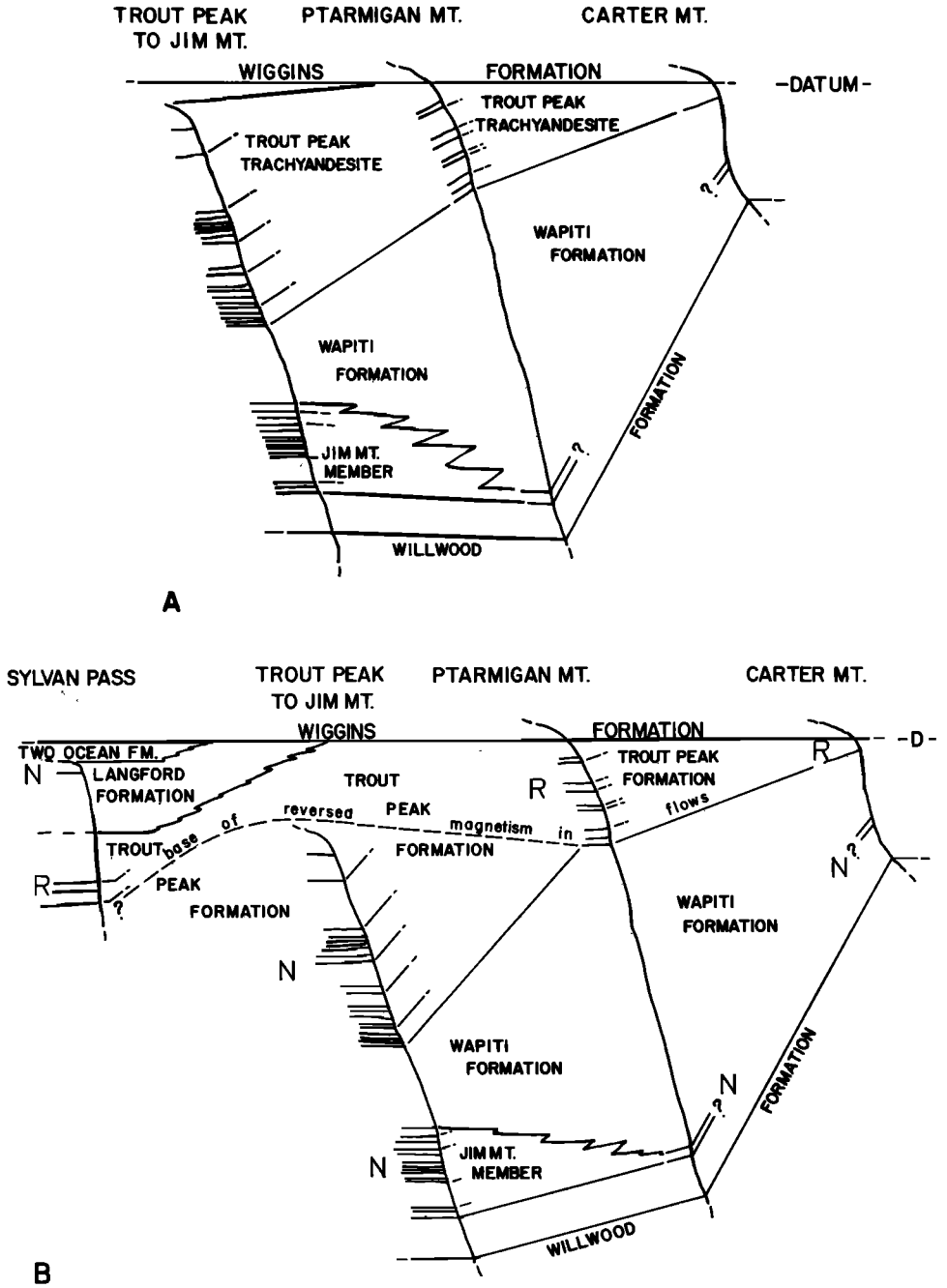



Fig. 4. (a) Correlation established by Nelson and Pierce (1968). (b) Revised correlation taking magnetic polarity into account.

shows anomalies 18-20 in the time interval between 45 and 50 m.y. B. P., but several authors [Sclater et al., 1974; R. L. Larson and Pitman 1975; Tarling and Mitchell, 1976] have pointed out inaccuracies in the extrapolation by which a polarity interval is assigned a radiometric age. These recent studies suggest that the Eocene anomalies are younger than was originally thought; the most significant readjustments show anomalies 22-24 occupying the time interval 45-50 m.y. B. P.

According to the studies cited above, the reversed polarity intervals between 45 and 50 m.y. B. P. range from about 0.1 to about 1.0 million years in length. If so, the entire sequence of reversed flows and associated sediments at Sylvan Pass, Ptarmigan Mountain, and Carter Mountain was formed during a period of time not greater than one million years in length and possibly considerably less. This analysis would be inaccurate if the flows were not between 45 and 50 m.y. B. P. in age, but we should note that no reversed interval in the Eocene (or in the entire Tertiary, for that matter) longer than three million years has yet been established. Another source of confusion is the possibility that the reversed sequence of flows could span an interval (or intervals) of normal polarity which are not observed because no volcanic activity occurred during that time. We feel, however, that the

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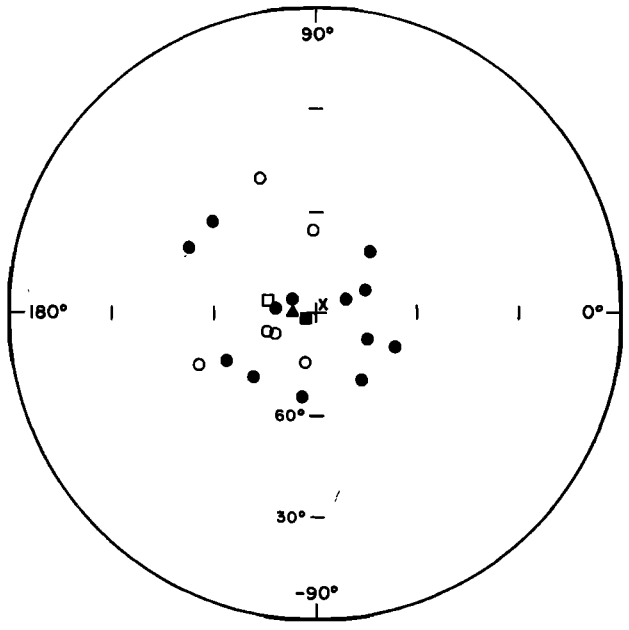



Fig. 5. Northern hemisphere equal area projection showing locations of pole positions.

possibility is very slight that a normal interval long enough to produce a linear anomaly (and thus be included in the time scale) would not be represented by any flows.

Of the original 34 flows sampled in this study, 15 were omitted from the calculation of a mean VGP either on the basis of a within-site $\alpha_{95}$ greater than 20° or because only two samples were usable. The remaining 19 poles are plotted in Figure 5. Average VGP positions have been plotted for the normal flows (13 in all), the reversed flows (6 flows), and the entire group of 19. Statistical parameters for these pole positions are given in Table 2. At the 95% confidence level the three VGP positions are not significantly different from each other or from the present rotational axis of the earth.

The mean VGP for these flows at 83.5°N, 177.4° E is a valuable addition to the few reliable pole determinations for North America from stable Eocene rocks. Of the half-dozen or so published pole positions, almost all must be rejected because the techniques used on the rocks studied are very likely to yield ambiguous results. Cox [1957] and Symons [1974] worked in areas near the continental margin which have almost certainly suffered serious tectonic deformation since the Eocene. Torreson et al. [1949] studied sediments which were not demagnetized. Strangway and McMahon [1973] studied very weakly magnetized kerogen shales which yielded an anomalous pole position. The Spanish Peaks dike swarm in Colorado, thought to be Eocene at the time of the study by E. E. Larson and Strangway [1969], has since been dated at about 25 m.y. B.P. [Stormer, 1972].

This leaves the work of Lovlie and Opdyke [1974] on some felsite and basic intrusives in Virginia, dated at 47 m.y. B.P. by Fullagar and Bottino [1969], as the only reliable study with which to compare our results. The VGP position calculated from these six sites is at 87.6° N, 45.9°E with an $\alpha_{95}$ of 12.0°. This pole position has been plotted in Figure 5 with the data from this study and is clearly not significantly different.

The results of both studies indicate that the Eocene VGP for North America was probably quite close to the present rotational pole. The Oligocene pole [Diehl et al., 1974] was apparently also close to the present pole, and the entire Upper Tertiary is believed to be a time of little apparent polar wander for North America [McElhinny, 1973, p. 203]. Pole positions for North American Cretaceous rocks, however, fall in the vicinity of the Gulf of Alaska [Gromme et al., 1967], so it would appear that the counterclockwise rotation required of North America since the Cretaceous must have been largely confined to the latest Cretaceous and the Paleocene. The high-latitude pole determined by E. E. Larson et al. [1969] from two basalt flows in Colorado dated at 59 m.y. B.P. may provide an even further restriction on the time of this motion, but it should be noted that a pole position based on two flows is almost certain to be strongly affected by secular variation.

These results have important implications for tectonic and paleomagnetic studies in the Pacific Northwest, where divergent pole positions of Eocene rocks are frequently observed [Cox, 1957; Beck, 1976; Cox and Simpson, 1976; Simpson and Cox, 1976]. If the geomagnetic field during the Eocene was well behaved and if the average pole position was near the present geographic pole, the divergent pole positions cannot be due to repeated and prolonged field excursions but must be due to large-scale tectonic rotations.

The total dispersion $\Delta_t$ for the entire group of pole positions is 25.4°. It is important to

TABLE 2. Average Pole Positions for Normal, Reversed and Total Flows

| | N | λ | θ | $\alpha_{95}$ | δ | κ |
|---|---|---|---|---|---|---|
| Normal flows | 13 | 86.7 | 202.9 | 12.3 | 25.6 | 10.0 |
| Reversed flows | 6 | 75.5 | 164.1 | 17.6 | 25.0 | 10.5 |
| Total | 19 | 83.5 | 177.4 | 10.1 | 25.4 | 10.2 |

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note that this value is not only a measure of dispersion due to secular variation of the pole $\Delta_p$ but includes a within-site dispersion term $\Delta_s$ caused by statistical variations in directions of the several samples from one flow and orientation errors. Even if there were no secular variation ($\Delta_p = 0$), the VGP positions would still show scatter due to $\Delta_s$. The dispersions are related by the expression $\Delta_t^2 = \Delta_p^2 + \Delta_s^2$, so in order to estimate the magnitude of secular variation during extrusion of the flows it is first necessary to make an estimate of $\delta_s$. This can be done because we have calculated $\delta_p$, the angular dispersion of the field direction, for each flow. From these values we can calculate a mean dispersion $\delta_m$ for the 21 flows involved in the calculation. This value is 14.0°. The value of $\Delta_s$ is then obtained from the field-to-pole mapping function for dispersion

$$\Delta_s^2 = \delta_m^2(5 + 18 \sin^2 \lambda + 9 \sin^4 \lambda)/8$$

where $\lambda$ is the site paleolatitude [Cox, 1970]. For $\lambda = 41°$ the value of $\Delta_s$ is 18.8°.

This results in a value of $\Delta_p$ of 17.0°. This estimate of paleosecular variation may be a bit high, as there is no way to consider other sources of between-site misorientation such as could be caused by local magnetic anomalies and by small uncorrected variations in the strike and dip of the different flows. A value of 17.0° is about 2° greater than that expected from model D of Cox [1969] using parameters for the recent field, but this difference is probably not significant. The value is also in agreement with observed paleosecular variation data presented by Creer [1962] and by Brock [1971].

C. S. Gromme (personal communication, 1976) pointed out that the six 'acceptable' reversed flows had greater dispersion of field directions (25.5°) and greater average within-site dispersion (18.2°) than the thirteen 'acceptable' normal flows (19.7° and 9.9°, respectively) and that this effect could be produced by a secondary CRM acquired in the normal Bruhnes field which would deflect a primary reversed TRM more than a primary normal TRM. However, most of this effect is probably due to the fact that (by coincidence) more samples were taken on the average from reversed flows. Because of this, reversed flows with 'acceptable' $\alpha_{95}$ values tend to have higher values of dispersion. When dispersion values for all reversed flows are tested against values for all normal flows, the difference is no longer significant.

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